MANIPAL
UNIVERSITY COLLEGE
MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

HISTOLOGY OF LYMPH NODE & SPLEEN

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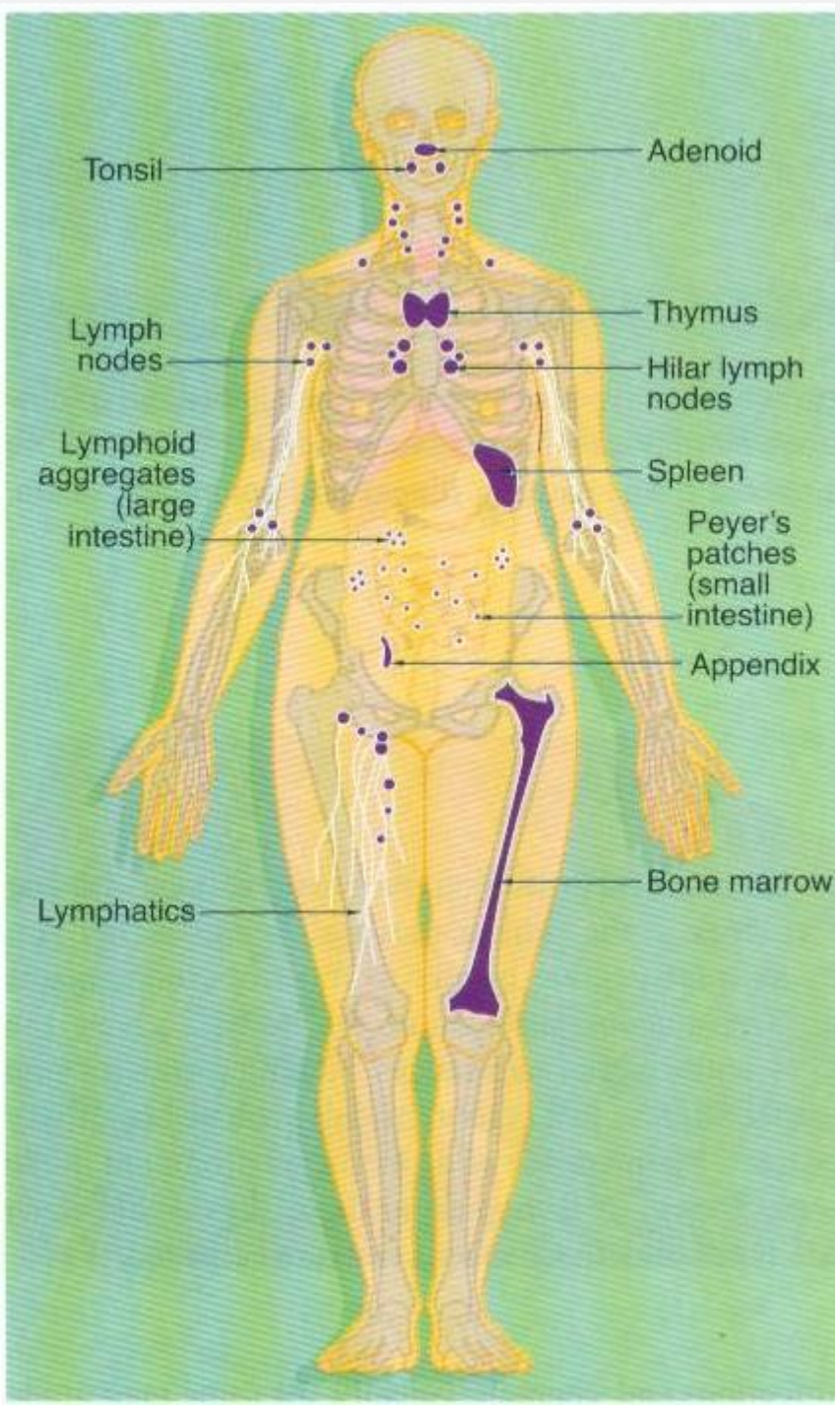


Only for instruction purpose..

#inspire

Learning objectives

- Explain and illustrate the microscopic structure of lymphoid tissue.


Tonsil
Adenoid
Thymus
Lymph nodes
Hilar lymph nodes
Lymphoid aggregates (large intestine)
Spleen
Peyer's patches (small intestine)
Appendix
Bone marrow
Lymphatics

Lymphoid organs - overview

A. Primary/Central lymphoid organs :

- Thymus
- Bone Marrow

Responsible for development & maturation of lymphocytes

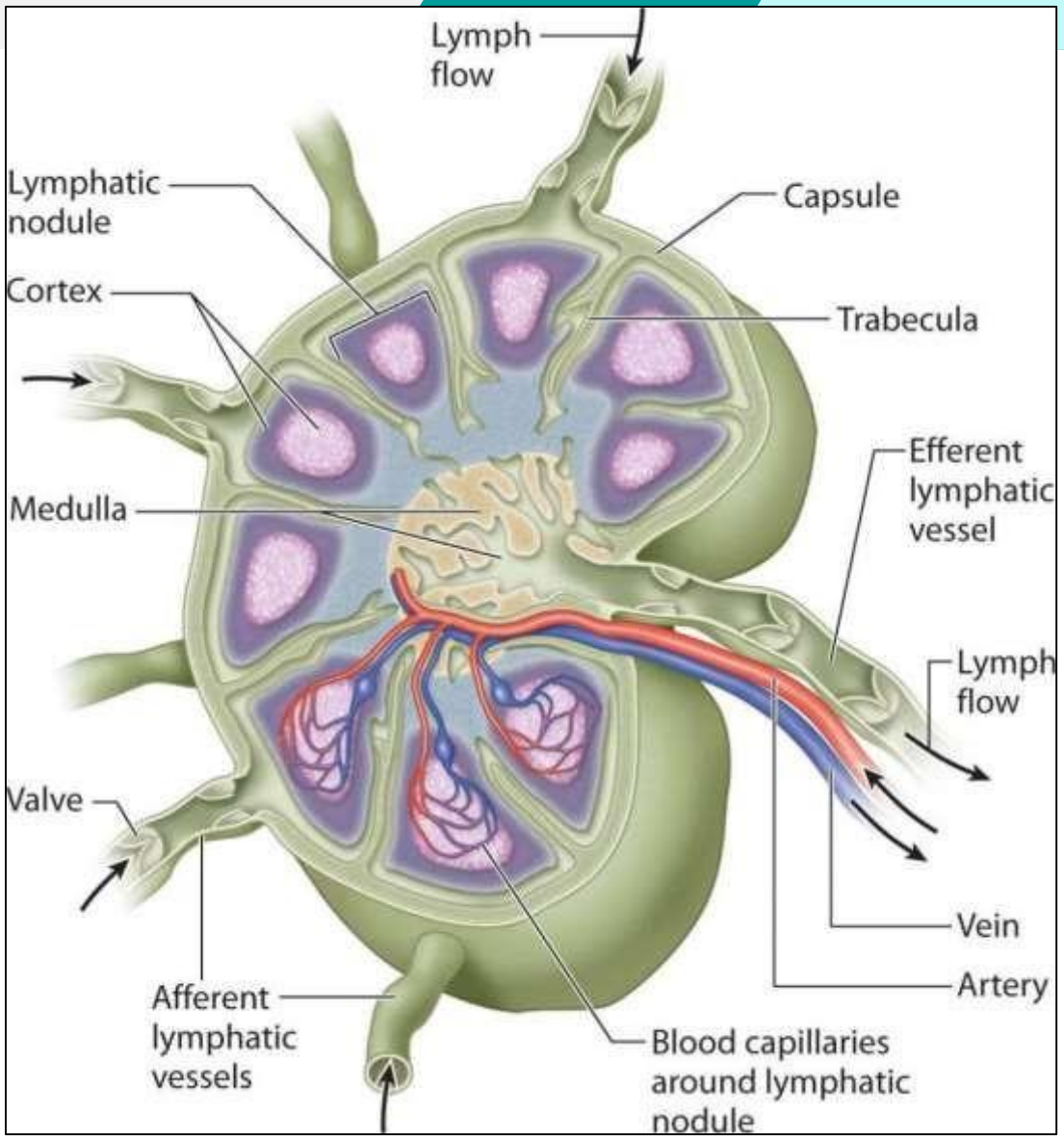
B. Secondary/Peripheral lymphoid organs :

- Lymph nodes
- Spleen
- Tonsils
- Other lymphatic tissues like MALT, EALT, SALT (solitary nodules, Peyer's Patches of ileum, Appendix)

Responsible for activation of lymphocytes

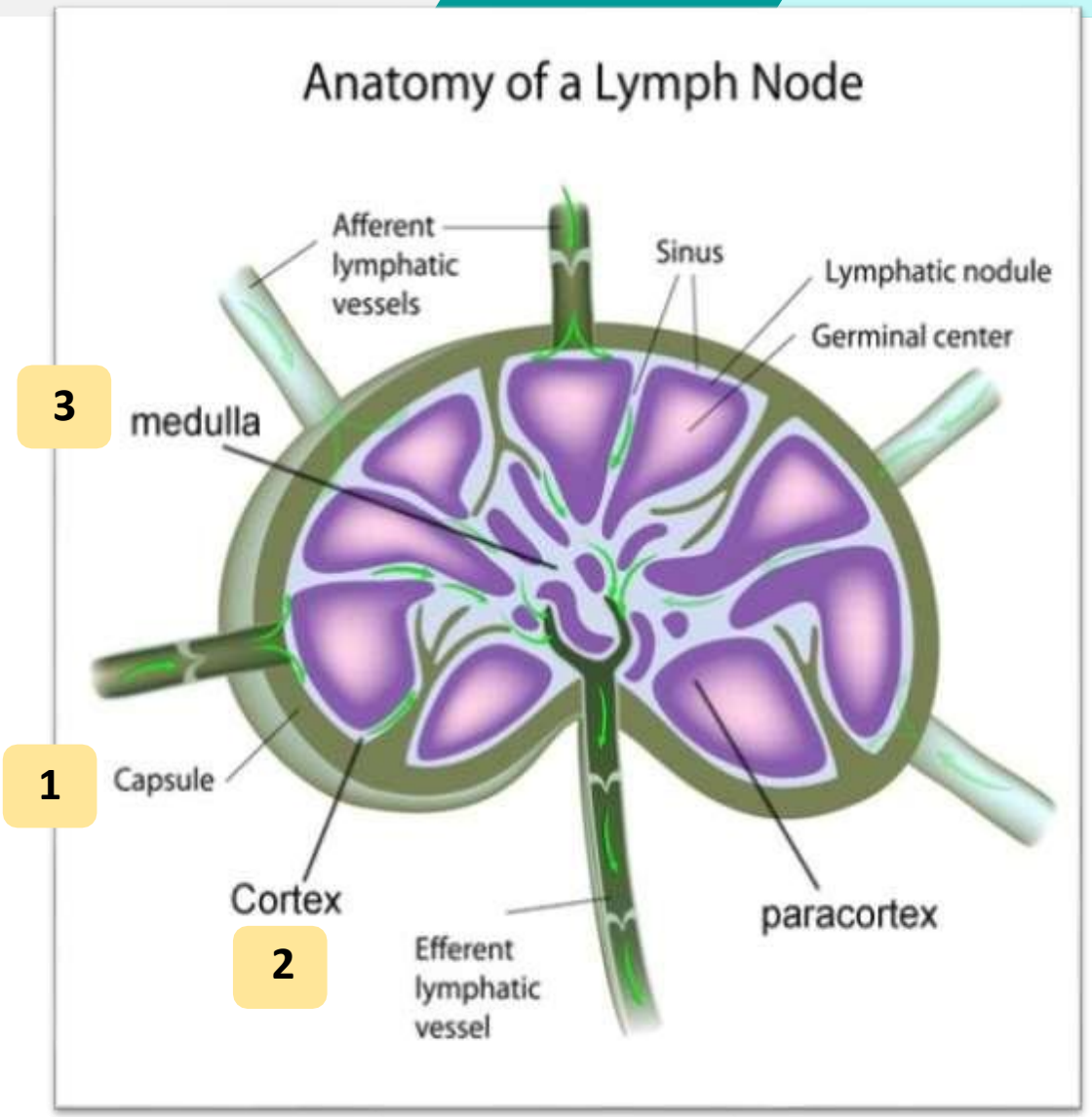
Lymph nodes

- Small solid masses of lymphoid tissue.
- Oval or reniform (kidney shaped) bodies.
- A slight depression on one side called hilum.



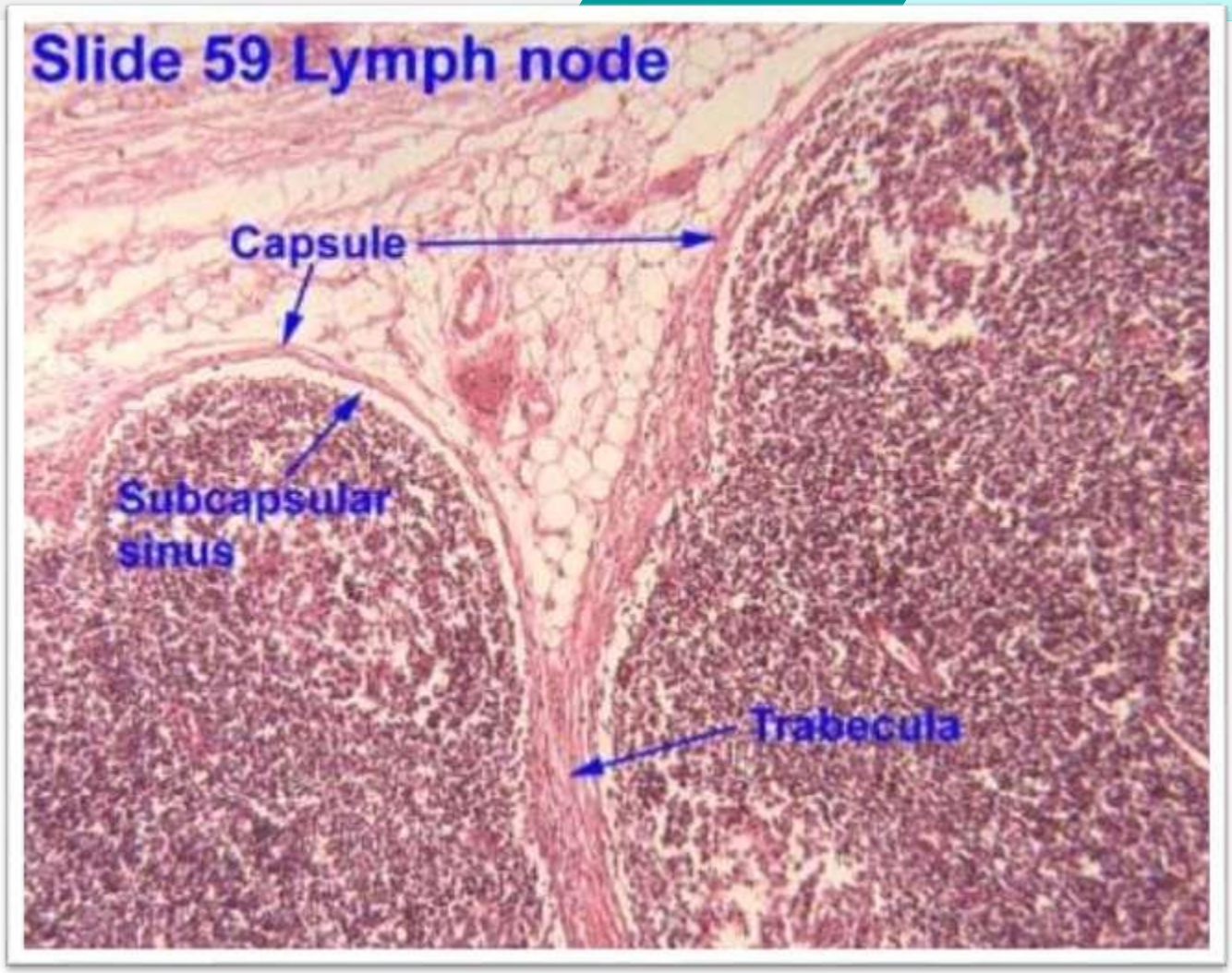
Microstructure of lymph node

1) Capsule
2) Cortex
3) Medulla



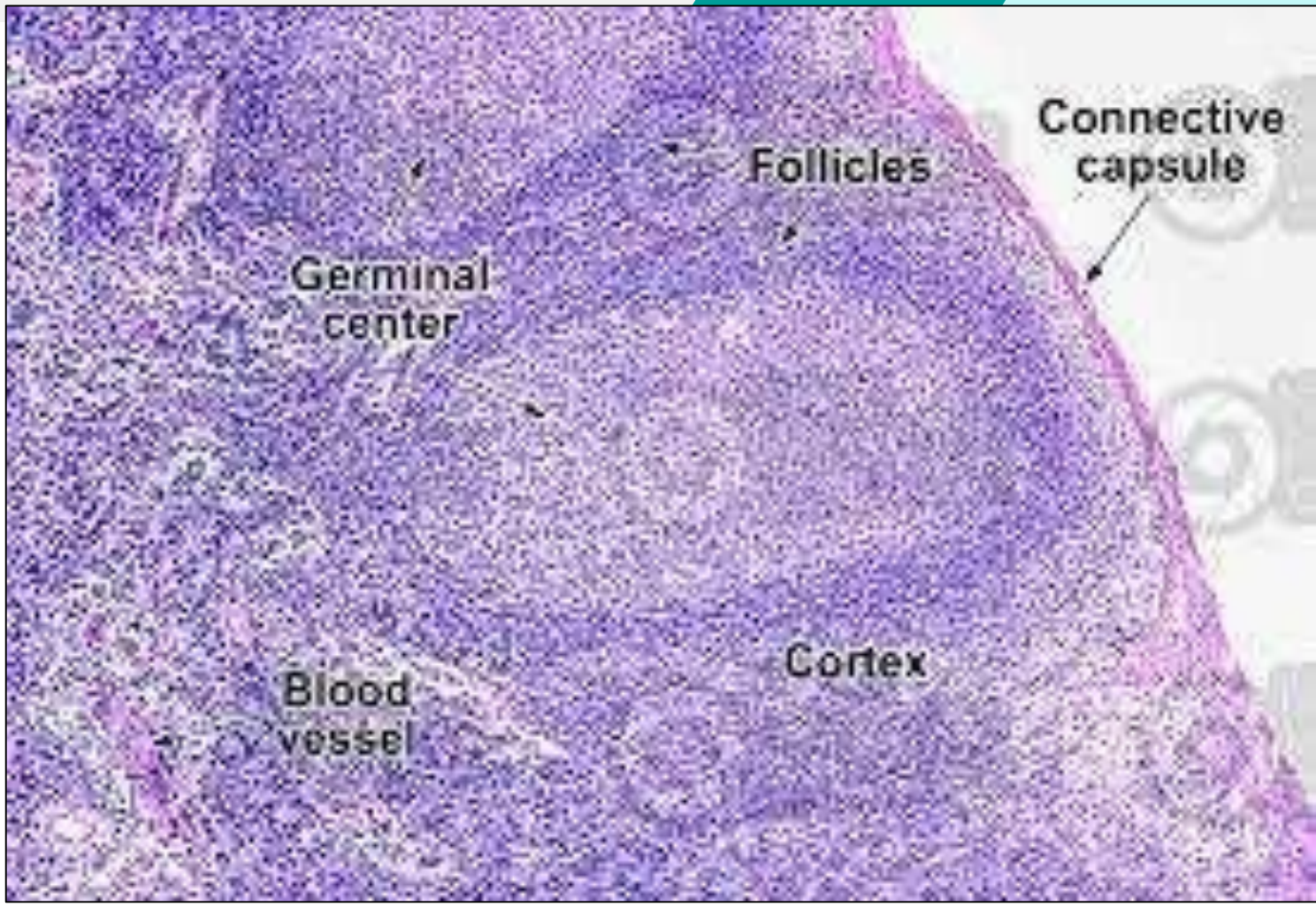
Capsule

- ✓ An outer fibrous covering composed of collagen fibres, few fibroblasts and elastic fibres.
- ✓ Sending septa (trabeculae) radially into the lymph node.
- ✓ Continuous with the network reticulin fibres- the reticulum.
- ✓ Subcapsular sinus beneath the capsule.



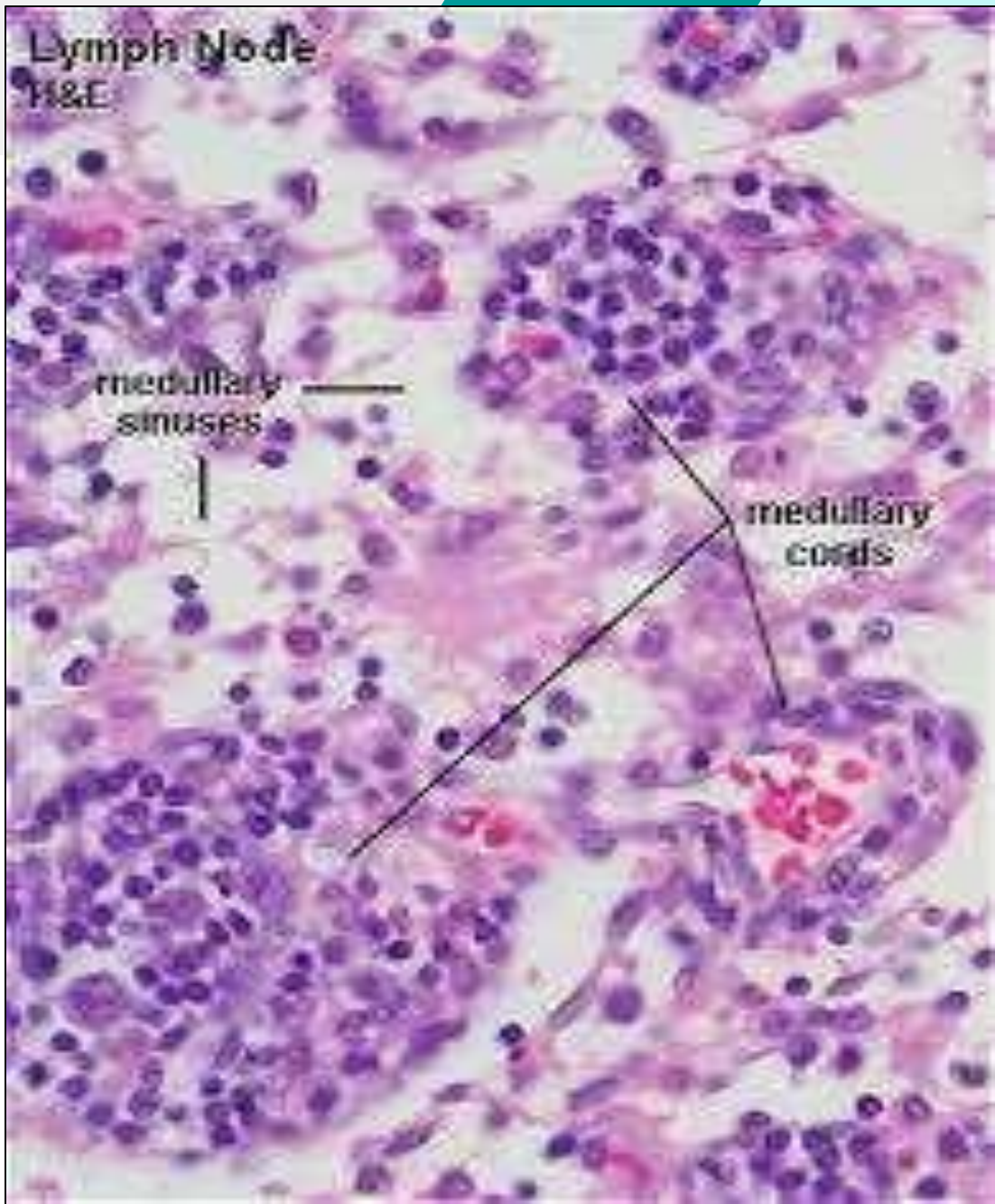
Cortex

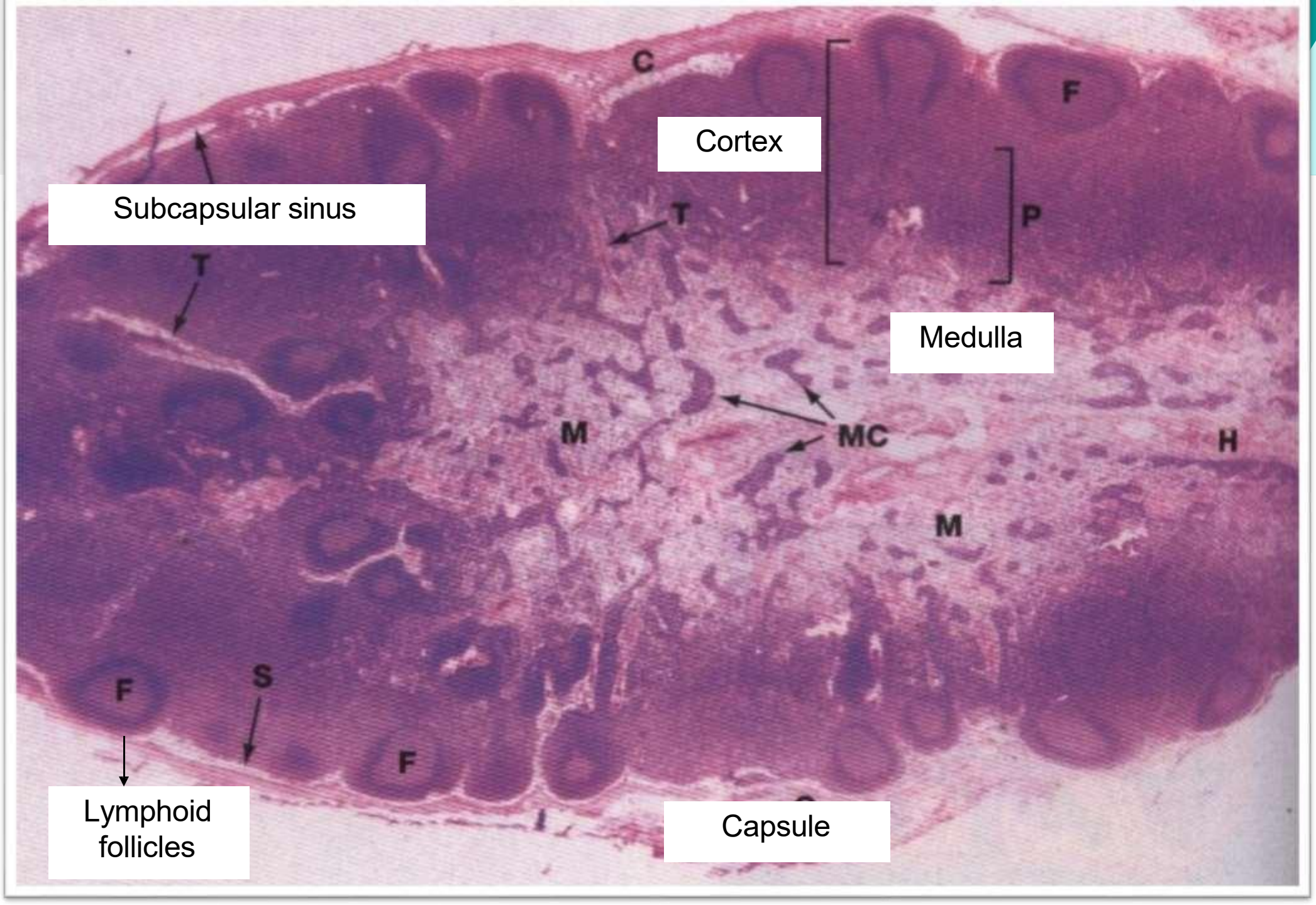
- ✓ Peripheral, contains densely packed cells in lymphatic follicles.
- ✓ Deficient at the hilum.
- ✓ The follicles:
  - ✓ **Germinal center**: Composed of lymphoblasts - larger, lightly staining & rapidly dividing cells.
  - ✓ **Mantle zone**: Composed of small mature lymphocytes which migrating from germinal centre.



Medulla

- ✓ Contains cavities filled with network of minute lymphatic channels and blood vessels.
- ✓ Lymphocytes form irregular branching medullary cords.
- ✓ Other cells: macrophages, plasmacytes and a few granulocytes.




C
F
Cortex
P
T
Subcapsular sinus
T
Medulla
M
MC
H
M
F
S
F
Lymphoid follicles
Capsule

Cells in lymph nodes

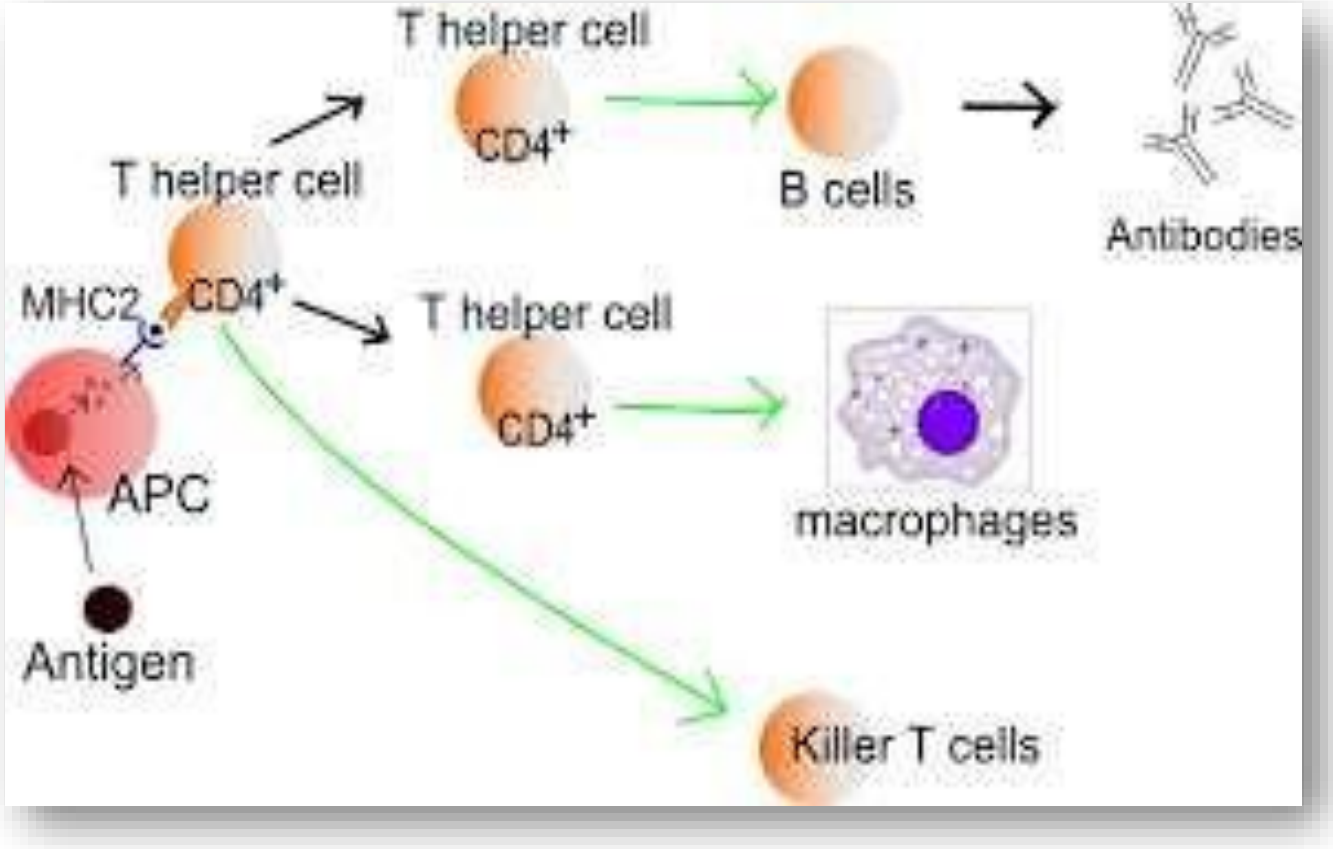
B lymphocytes:

- Synthesize antibodies of the IgA occurring in alimentary secretions and in IgE (homocytotropic antibody mainly related to mast cell activities).

T lymphocytes:

- CD4 + helper T cells stimulating B-cell activity.

- CD8 + cytotoxic T cells engaging in the destruction of virus- and parasite-infected cells, and of neoplastic cells.



Functions of lymph nodes

i. The **exposure** of **foreign material** in the lymph to macrophages.

ii. The **trapping of antigens** by different mononuclear phagocytes.

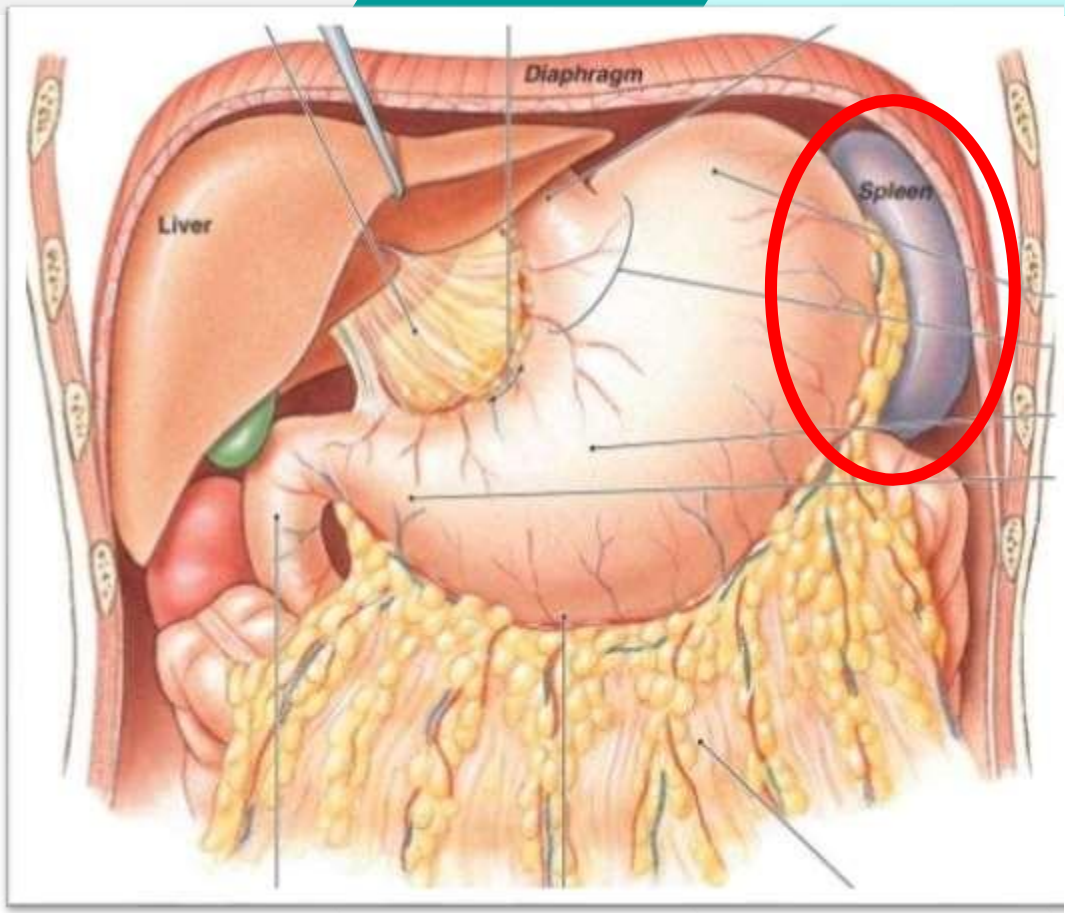
iii. **Production** of **lymphocytes** and a pool of **stem cells** - antibody-producing B lymphocytes or mature T lymphocytes.

iv. **Produce** both cell-mediated and humoral **immune response.** (antibody production)

Spleen -Introduction

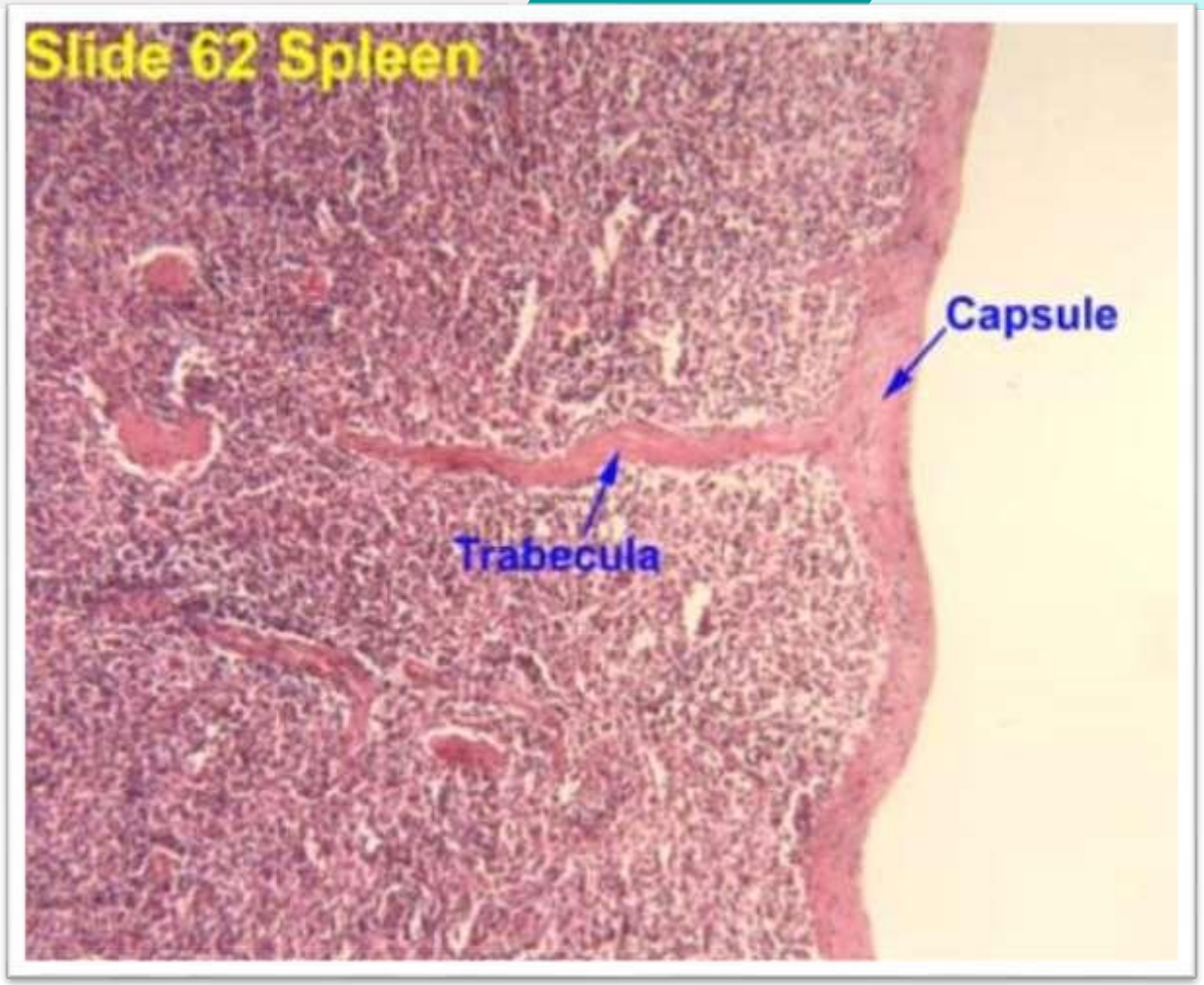
- A large encapsulated mass of haemo-lymphoid tissue.

- Position: Left hypochondrium wedged between the fundus of the stomach and the diaphragm.



Fibrous framework of spleen

- **Outer covering: Serosa** covers the entire organ except at its hilum.
- **Capsule**: a continuous layer about 1.5 mm thick, rich in collagen.
- **Trabeculae**: composed of dense irregular connective tissue, rich in collagen and elastic fibres.
- Numerous trabeculae extend inside and branch profusely to form a supportive framework.

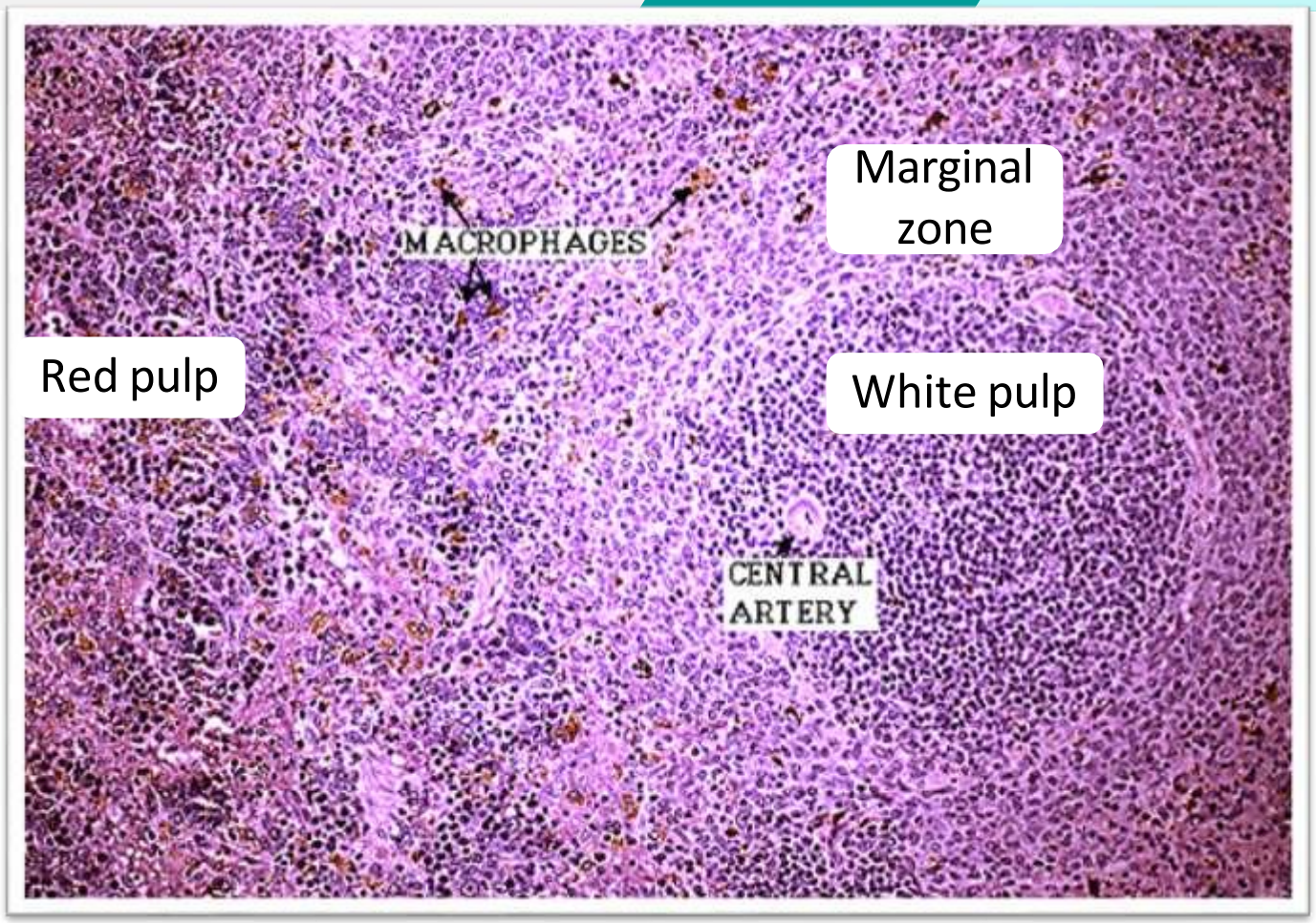


Microstructure of spleen

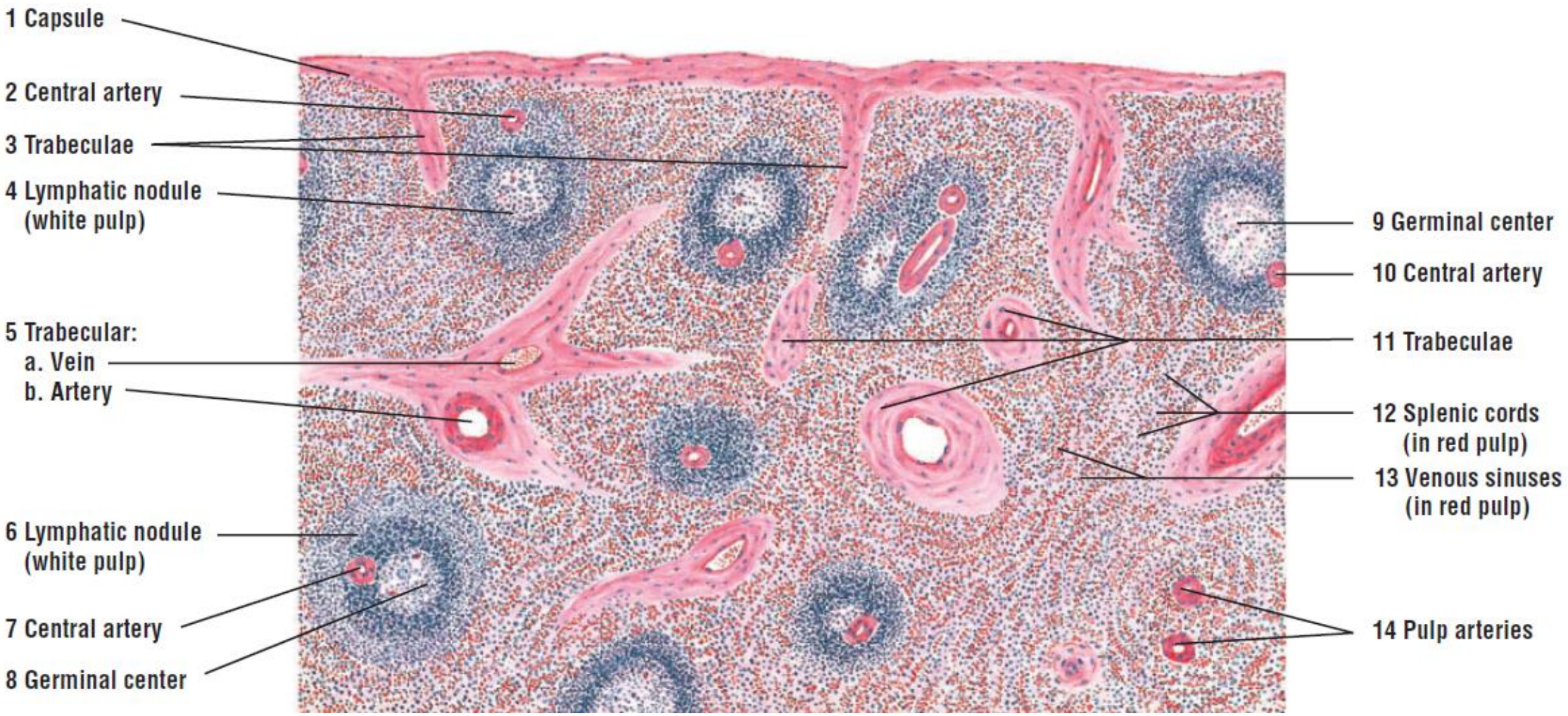
I. **White pulp**- Lymphoid tissue in which B and T lymphocytes can mature and proliferate.

II. **Red pulp**- A complex system of interconnected spaces (sinusoids) having large numbers of phagocytic macrophages.

III. **Marginal zone-** Narrow area at the junction of white and red pulp.



I. White pulp


1 Capsule
2 Central artery
3 Trabeculae
4 Lymphatic nodule (white pulp)
5 Trabecular:
a. Vein
b. Artery
6 Lymphatic nodule (white pulp)
7 Central artery
8 Germinal center
9 Germinal center
10 Central artery
11 Trabeculae
12 Splenic cords (in red pulp)
13 Venous sinuses (in red pulp)
14 Pulp arteries

White pulp

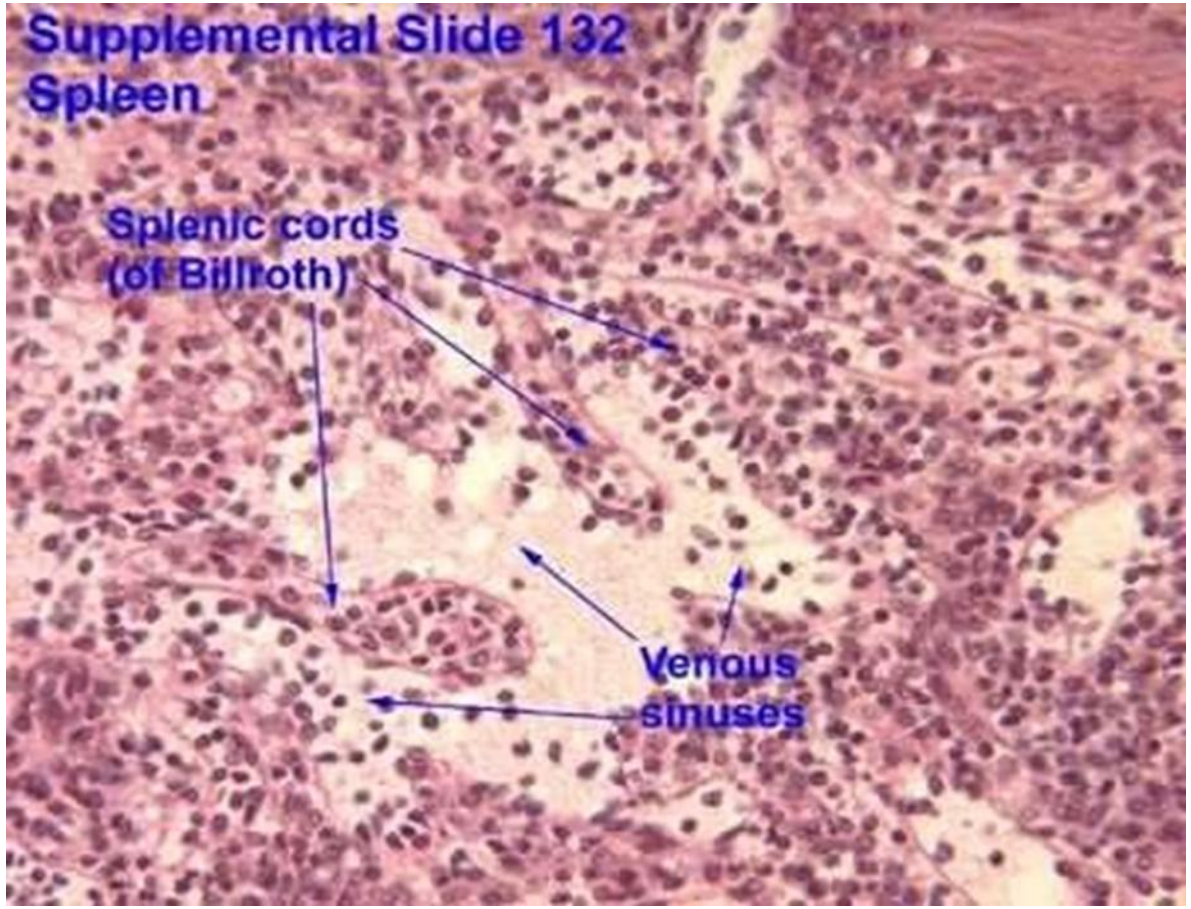
- white pulp is made up of lymphocytes that surround arterioles, it is in the form of cord-like aggregations of lymphocytes that follow the branching pattern of the arterioles. The cords appear to be circular in transverse section.
- At places, the cords are thicker than elsewhere and contain lymphatic nodules similar to those seen in lymph nodes. These nodules are called ***Malpighian bodies***.
- Each nodule has a germinal centre and a surrounding cuff of densely packed lymphocytes.
- The nodules are easily distinguished from those of lymph nodes because of the presence of an arteriole in each of them.

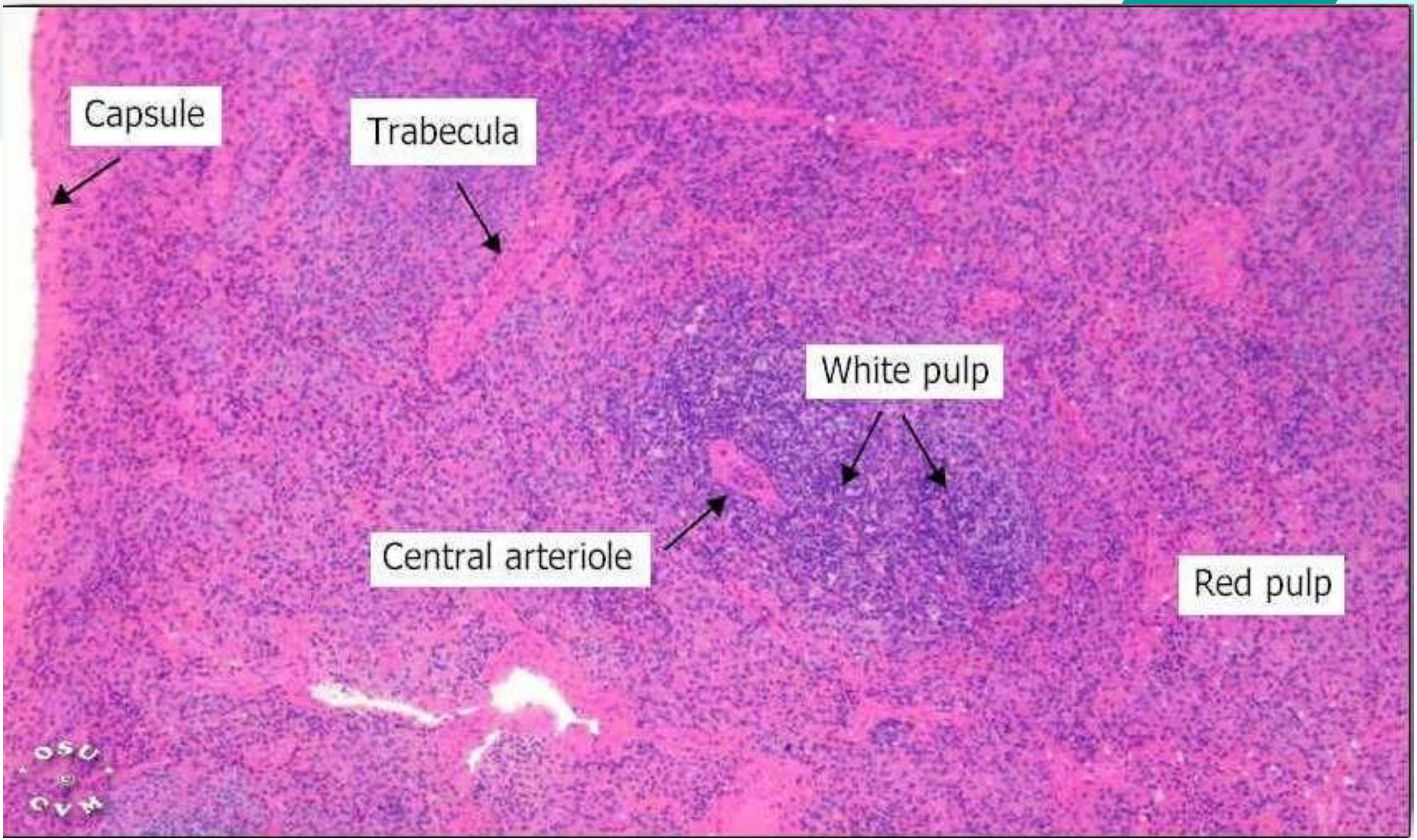
White pulp

- The arteriole is placed eccentrically at the margin of the germinal centre (between it and the surrounding cuff of densely packed cells).
- **Periarteriolar lymphoid sheaths (PALS)** are specialized, T-cell-rich cylindrical structures within the white pulp of the spleen that surround central arteries
- The functional significance of the white pulp is similar to that of cortical tissue of lymph nodes.
- Most of the lymphocytes in white pulp are T-lymphocytes.
- Lymphatic nodules of the white pulp are aggregations of B-lymphocytes.
- The germinal centres are areas where B-lymphocytes are dividing.

Red pulp

- The red pulp is like a sponge. It is permeated by spaces lined by reticular cells.
- The intervals between the spaces are filled by B-lymphocytes as well as T-lymphocytes, macrophages, and blood cells.
- These cells appear to be arranged as cords (splenic cords, of Billroth). The cords form a network.



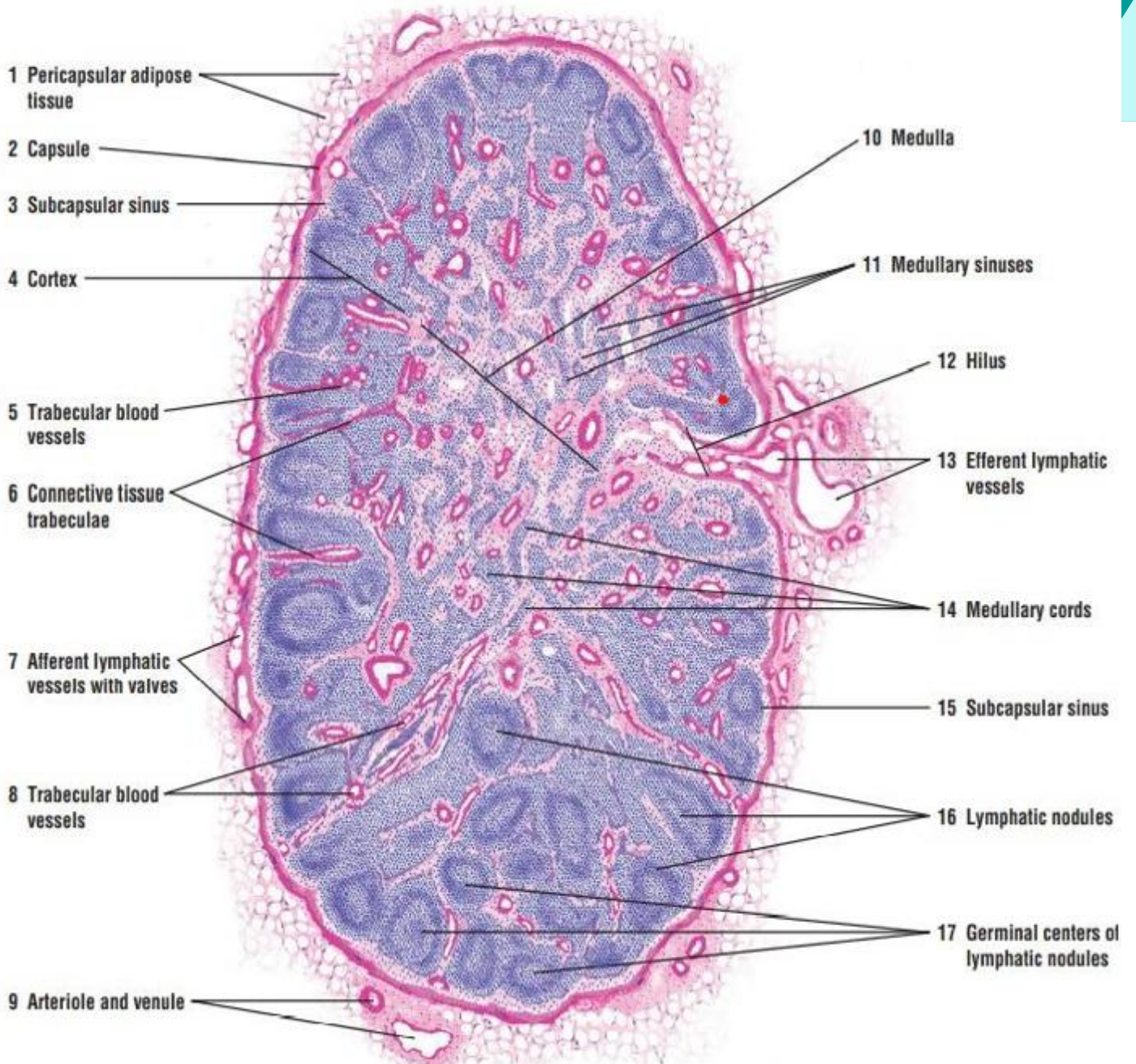

Capsule
Trabecula
White pulp
Central arteriole
Red pulp

Functions of spleen

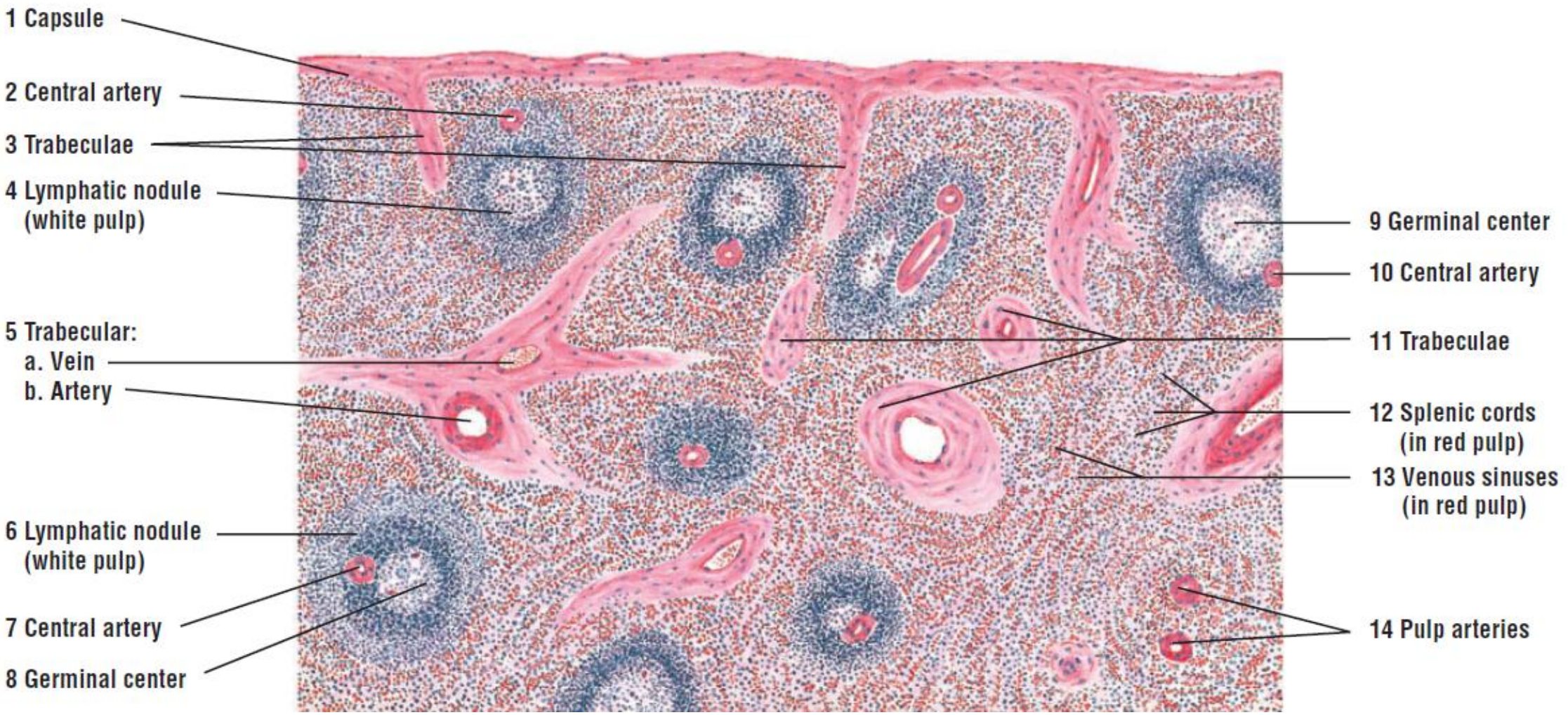
i. **Removal** of **particulate material** including ageing erythrocytes from the circulation.

ii. **Provision** of **lymphocytes and antibodies** as part of the body's system of secondary lymphoid tissues.

Lymph node to draw



Spleen to draw



THANK YOU !